

SECTION 31 23 23
FILL, BACKFILL AND COMPACTION (EMEA)

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PART 0 PROJECT-SPECIFIC AMENDMENTS

The following amendments to this Section apply only to this Project:

- 1.3.C.1a: Remove ‘-12’
- 1.4.A: Insert ‘in 31 23 23 Fill, Backfill and Compaction’
- 1.5.A: Insert ‘(third party)’
- 1.5.C.5: Replace ‘LAB’ with ‘EN ISO 17892 Geotechnical investigation and testing’
- 1.6.B: Add ‘In-situ site testing results to be reviewed and updates made to the Ground Improvement Plan (Ref: STB1A-1.0-S-1000-ALL) if necessary.’
- 2.3.A: Remove ‘or “Flowable Fill”
- 2.5.B: Remove this paragraph in its entirety.
- 2.6.B: Remove this paragraph in its entirety.
- 2.8: Add new Section ‘GROUND IMPROVEMENT – LIME STABILISATION’
- 3.1.A: Delete ‘fly ash or’
- 3.1.B-E: Add new paragraphs on lime stabilisation
- 3.1.F: Add ‘soil-lime mixture / ‘ and ‘The soil-lime mixture should be left to set for 24-48 hours to allow for chemical reactions to take place. The mixture / suitable fill should then be compacted in layers to 95% of the Modified Proctor Density (EN 13286-2)’
- 3.1.G-H: Add new paragraphs on lime stabilisation
- 3.1.I: Remove paragraph in its entirety
- 3.2.A: Replace “Suitable Material” with “Lime stabilised”
- 3.2.E: Remove ‘Place “Flowable Fill” in layers not exceeding 900 mm in depth.’
- 3.3.E: Delete this paragraph in its entirety
- 3.3.F: Insert ‘using fill material as per details. The material’
- 3.4.D: Replace ‘below final grade’ with ‘above the utility asset’
- 3.4.E: Delete this paragraph in its entirety
- 3.5.B: Remove ‘-1’

PART 1 GENERAL

1.1 SUMMARY

- A. Section Includes: Materials and requirements for site filling, backfilling against structures, pipe bedding and trench backfilling.

1.2 RELATED REQUIREMENTS

- A. 03 30 00 - Cast-in-Place Concrete.
- B. 31 23 16 - Structure Excavation.
- C. 31 23 16.13 - Trenching For Site Utilities.

1.3 SUBMITTALS

- A. Product Data: Flowable fill mix designs shall be clearly labeled as to intended locations of use. Comply with Section 03 30 00 - Cast-in-Place Concrete.
- B. Certification of "Suitable Material" and "Granular Material," and pipe bedding to be submitted for approval at least 10 days before placing material.
- C. Quality Control Procedures:
- 1. Test Reports:
 - a. Tests in accordance with EN ISO 17892-12.
 - b. Particle size analysis test results, in accordance with EN 933, for proposed porous fill, granular under slab fill, and pipe bedding.
 - c. Particle size analysis test results in accordance with EN 933, and plasticity index test results in accordance with EN 16994, for proposed "Suitable Material."
 - 2. Log Data: On a weekly basis, giving information on area worked (locate by Project Site reference grid), nature of fill placed (material used, method of compaction, moistening or aeration), weather conditions, type of equipment used, and other relevant data.

1.4 DEFINITIONS

- A. Refer to Table 312323/A in 31 23 23 Fill, Backfill and Compaction

1.5 QUALITY ASSURANCE

- A. Services of an Inspection and Testing Agency (third party) will be engaged to perform inspection of Work and make soil compaction tests.
- B. Referenced Codes and Standards:
- 1. All earthworks shall comply with the requirements of EN 1997.
 - 2. The products, works or services to be procured are generally described by reference to European Standards and relevant national annexes.
 - 3. When a Standard, Specification or Code of Practice is referred to, then the latest edition of such Standard, Specification or Code of Practice is implied.
- C. Extent of Testing

1. The schedule of tests indicated below (Table 312323/A) is non-exhaustive. Other tests within reason may be required to prove the specified product or material has been installed to the relevant standards.
2. The Contractor shall incorporate in the schedule of tests as a minimum the tests detailed in the following table together with all additional tests required by the Contract. The Contractor is responsible for arranging and bearing the cost of carrying out the any required tests.
3. The Contractor is to ensure testing is carried out in accordance with EN 1997.
4. Tests comparable to those specified in this Section will be necessary for any equivalent work, goods or materials proposed by the Contractor.
5. (L) indicates that a certified ~~LAB~~ EN ISO 17892 Geotechnical investigation and testing test report or certificate is required.
6. (*) indicate a manufacturer's test or certificate if required.
7. Unless otherwise shown in this Section test certificates for work, goods or materials as scheduled under any one Section/Clause are required for all such work, goods or materials in the Works.
8. As soon as possible after the Contract has been awarded, the Contractor shall supply and deliver to a laboratory nominated by the A/E of Record such samples of materials used in the works as the A/E of Record may request.
9. These samples are to be provided free of charge and are additional to the quality control testing to be undertaken by the Contractor as required.
10. The Contractor shall develop and maintain a comprehensive schedule of tests for all works. The results of all tests are to be collated and presented by the Contractor. As a minimum the date of test, position of test including a drawing, result and whether the test has passed or failed should be identified. Any failed tests should be brought to the attention of the A/E of Record as soon as they become available. Test results should be returned and be accepted by the A/E of Record before dependent Works are progressed.

D. Schedule of Quality Control Testing

1. The following testing, set out in Table 312323/A, to be carried out by the Contractor shall be included as a minimum and shall be added to as necessary for the execution and completion of the Works, unless otherwise agreed by the Employer's Representative.
2. The scope of the testing covered in Table 312323/A should not be regarded as exhaustive. Routine tests carried out by manufacturers and suppliers in compliance with an EN or other standard or specification are not included but where a standard or specification makes provision for a test certificate this is indicated in the table.
3. Where the quantity of material used in the Works is less than any quantity described under "Frequency of Testing" then the quantity described under "Frequency of Testing" shall be read as the quantity used in the Works.
4. Details of material acceptability and compaction tests to be carried out by the Contractor are given in the following table.
5. The Contractor shall carry out CBR or other approved tests in accordance with EN 1997 and associated EN standards to demonstrate subgrade strength. The locations of the tests are to be agreed in advance with the Employer's Representative.
6. Should alternative materials or products be used to those included in the Work Requirements, additional tests may be required.
7. The Contractor shall record and maintain drawing showing the locations of all testing carried out on site to be submitted along with the test results.
8. This document uses the following abbreviations.

L:	Laboratory accreditation granted by the National Accreditation Board
MCV:	Moisture Condition Value
CBR:	California Bearing Ratio
DCP:	Dynamic Cone Penetration
PL:	Plasticity Limit
MC:	Moisture Content
OMC:	Optimum Moisture Content
PLT:	Plate Load Test
NDG:	Nuclear Density Gauge
SRD:	Sand Replacement
SMC:	Saturation Moisture Content

1.6 IN-SITU SITE TESTING

- A. In addition to the requirements described earlier, the Contractor will undertake CBR testing of the exposed formation level under all roadways and external hardstanding areas. CBR tests to be at 50 m centers along roadways, and a minimum of 1 per 100 m² for external hardstanding areas, or otherwise directed by the A/E of Record. Alternative methods to CBR testing, such as Cone Penetrometer Tests or CPTs, may be carried out with the advance approval of the A/E of Record where local conditions dictate.

- B. In-situ site testing results to be reviewed and updates made to the Ground Improvement Plan (Ref: STB1A-1.0-S-1000-ALL) if necessary.

PART 2 PRODUCTS

2.1 MATERIAL AVAILABILITY

- A. Reuse of site won material will be considered on Project Site and from Project excavation but quality and quantity are not guaranteed for all uses. Review Geotechnical Report and Project Site to make independent judgments regarding materials available. Should quality be unacceptable or quantity insufficient for intended use, obtain material of required quality and quantity from off Project Site sources.
 - 1. Suitable Material.
 - 2. Granular Material.
- B. Off Project Site material of each type shall be from a single source if possible. Request inspection of imported materials area and obtain approval of materials proposed for use.

2.2 SITE FILL

- A. Lawn and Planting Areas and Other Non-Load Bearing Areas: according to standard details as defined.
- B. Load Bearing Areas:
 - 1. For fills under slabs-on-grade, according to the A/E of Record's design as defined.
 - 2. For fills under footings, according to the A/E of Record's design or "Flowable Fill," as defined.

2.3 BUILDING FOUNDATION BACKFILL

- A. According to the A/E of Record's design or "~~Flowable Fill~~", as defined.

2.4 POROUS FILL

- A. For drainage course under slabs-on-grade and for bedding and backfill for sub-drainage piping beneath slabs-on-grade and at foundation walls, use coarse aggregate sizes in accordance with EN 1295, EN 933, and EN 1744.
- B. Refer to standard details for selected backfill.

2.5 UTILITY TRENCH PIPE BEDDING

- A. Refer to standard details for more information on bedding and surround.
- ~~B. "Flowable Fill" may be used with the approval of the Project Manager for bedding and surround as indicated and/or as an alternate to the specified material.~~

2.6 UTILITY TRENCH BACKFILL

- A. Trenches under project site paving, slab-on-grade, or any other structure according to standard details.
- ~~B. "Flowable Fill" may be used with the approval of the Project Manager for trench backfill as indicated and/or as an alternate to the specified material.~~

2.7 WATER

- A. Provide water and transport water to and on Project Site, as required to ensure proper moisture content for compaction.

2.8 GROUND IMPROVEMENT – LIME STABILISATION

- A. Lime
 - 1. Hydrated lime or quicklime conforming to EN 459-1 (Building Lime) specifications.
 - 2. Lime must be free of contaminants and delivered in sealed bags or tankers with proper labelling.
- B. Water
 - 1. Water used for mixing should meet the quality requirements of ÖNORM B 5014.

PART 3 EXECUTION

3.1 PROJECT SITE FILLING

- A. After stripping of topsoil, areas to receive fill shall be proof-rolled under observation of Soils Inspection and Testing Agency. Any area that, in the opinion of the Agency, ruts or "pumps" excessively shall be excavated and backfilled with "Suitable Material" compacted to 98 percent of Standard Proctor maximum dry density as directed by Soils Inspection and Testing Agency. As an alternative, the unstable soil may be chemically stabilized by adding ~~fly ash~~ or lime if approved by Soils Inspection and Testing Agency and the Owner's Representative.
- B. Unstable / Unsuitable materials must be removed to a depth specified in project documentation. The moisture content of the soil must be verified, and adjusted, if necessary, to reach optimal levels for mixing. The lime content required depends on the soil characteristics and the design strength requirements. Laboratory tests, such as the Eades and Grim pH test and Unconfined Compressive Strength (UCS) testing, should guide the final lime content.

General recommendations for heavy-loaded conditions are as follows:

- Lime Dosage: 5-8% of the dry weight of soil.

- Higher lime content (~7-8%) is recommended for clay-rich, high-plasticity soils to ensure stability under heavy traffic loads.
- Determined through laboratory testing, such as:
 - pH Testing (Eades and Grim method): Target pH > 12.4 for adequate lime-soil reaction.
 - Unconfined Compressive Strength (UCS): Target strength of 1.5-2.5 MPa at 28 days.

C. Depth of Lime Treatment

- Heavy Traffic Roads: 0.4 - 0.5 meters.
- Access Roads or Secondary Roads: 0.3 - 0.4 meters.
- Transition Zones (e.g., embankments): Increase depth up to 0.6 meters for enhanced stability.
- Lime Application:
 - Lime must be spread uniformly using calibrated spreaders.
 - Lime must be incorporated into the soil using rotary mixers, ensuring thorough mixing to a depth of 0.3-0.6 meters.
- Lime Quantity:

The lime requirement depends on laboratory testing but typically ranges from 5% to 8% of the dry weight of soil:

- For 1 m³ of soil weighing approximately 1,800 kg (dry density), lime requirement = 90-144 kg/m³.
- For a typical 0.5 m depth treatment: Lime requirement = 45-72 kg per square meter.
- The optimal lime content (2-8% by weight of dry soil) required to achieve the desired Unconfined Compressive Strength (UCS) is to be determined as per ÖNORM B 4400 guidelines.
- Lime-soil mixes are to be tested for UCS, California Bearing Ratio (CBR), and durability under freeze-thaw and wet-dry cycles.
- Lime Treatment Coverage:

The coverage percentage for lime treatment depends on the area and uniformity of load distribution:

- Roadways and Traffic Zones:
 - Full lime treatment coverage (100%) is required for the entire width of the roadway.
 - Extend treatment at least 1 meter beyond the edge of the pavement to ensure lateral support.
- Under Heavy-Loaded Slabs and Pavements:
 - Lime stabilization should cover the entire foundation footprint.
 - Extend treatment 0.5-1 meter beyond the slab edges for stability and to prevent differential settlement.
- Lower Stabilization Percentage
 - 20% stabilization: Suitable for surface improvement and light structural applications, such as temporary roads or minor embankments.
 - 50% stabilization: Designed for heavier structural loads, permanent roadways, and flood protection embankments.

D. Lime Application

- Spread lime uniformly across the soil using a calibrated spreader.
- Lime type: Hydrated lime (Ca(OH)₂) or quicklime (CaO), depending on availability and contractor preference.

E. Mixing

- Rotary mixers to blend lime and soil to the specified depth (e.g., 0.4-0.5 meters).
- Complete and uniform mixing to avoid untreated soil pockets.

F. Place soil-lime mixture / suitable fill material in horizontal layers not exceeding 200 mm in loose depth. The soil-lime mixture should be left to set for 24-48 hours to allow for chemical reactions to take place. The mixture / suitable fill should then be compacted in layers to 95% of the Modified Proctor Density (EN 13286-2), and compact each layer with appropriate plant as noted in Table on page 22. Except as otherwise approved, each layer of material shall extend the entire length and width of the area being filled. Do not place material on surfaces that are muddy, frozen or which contain frost. Do not place frozen fill.

G. Lime stabilised areas are to cure for a period of 7 -14 days maintaining moisture using light water spraying or covering the treated area with geotextile/plastic sheeting.

H. Monitoring and Quality Control:

- Field Testing:

- In-situ density tests (e.g., nuclear density gauge, sand cone test).
- UCS tests on samples extracted post-compaction.
- Laboratory Testing:
 - Conduct Atterberg limits, pH tests, and strength tests before and after stabilization.
- During Construction
 - Monitor pH and temperature during mixing to confirm proper chemical reactions.
 - Verify lime spread rates using in-situ sampling.
 - Perform density and strength tests at regular intervals to ensure compliance.
- Post-Construction
 - Conduct post-curing UCS tests on core samples to confirm the target strength.
 - Inspect treated areas for signs of cracking, settlement, or uneven compaction.

~~I. Place “Flowable Fill” under footings in layers not exceeding 900 mm in depth. Do not place material on surfaces that are muddy, frozen or which contain frost. Placement shall be formed such that fill extends no greater than 600 mm beyond footing.~~

3.2 BUILDING FOUNDATION BACKFILL

- A. Structures have been designed for backfill pressure from ~~“Suitable Material”~~ “Lime stabilised” fill. Protect against and repair damages caused by pressure from accidental loading.
- B. Before placing backfill against structures, provide the following:
 - 1. Permanent or temporary horizontal bracing of structures to be backfilled.
 - 2. Cut off sheeting to depth permitted by local governing authority.
 - 3. Remove trash and debris from excavations.
- C. Allow sufficient time for completion of:
 - 1. Curing of structures until they are sufficient to withstand backfilling pressures.
 - 2. Bracing slabs.
 - 3. Work specified in other Sections which is applied to or is adjacent to structures to be backfilled, and the inspection and approval of such Work.
 - 4. Adjacent underground Utility Work specified in other Sections, and inspection, approval, and recording of locations of Utility Work.
 - 5. Removal of formwork.
- D. Backfill by methods which prevent damage to structures. Where backfill occurs on both sides of a structure, place and compact each lift simultaneously on each side. Where backfill occurs on more than one (1) side of building, place and compact each lift to same level at each side. Slope top of backfill away from structures.
- E. Place “Granular Material” in horizontal layers not exceeding 200 mm in loose depth. ~~Place “Flowable Fill” in layers not exceeding 900 mm in depth.~~

3.3 UTILITY TRENCH PIPE BEDDING

- A. Place bedding material under, around, and above top of pipe in 150 mm layers where required by the details in the project drawings. Compact by rodding or spading, so as not to damage the pipe or coating.
- B. Place bedding to firmly support pipe along its entire length where required by details in the project drawings.
- C. Bring bedding evenly on both sides of the pipe.
- D. Where soil at the bottom of the trench is unsuitable, remove unsuitable material and stabilize the trench bottom with a granular material similar to bedding material except that maximum size shall be 50 mm. Provide depth of stabilization as required to construct a firm subgrade for bedding material including potential use of geogrid wrapping of fill to further strengthen where required. Provide concrete cradles where required to bridge highly unstable soils and any other pipe when clear space between pipes is less than 300 mm.
- ~~E. Place “Flowable Fill” bedding in lifts so as to avoid floating or misalignment of pipe and to provide lateral support along pipe sides. Allow “Flowable Fill” to harden between lifts.~~
- F. The bed, haunch and surround using fill material as per details. The material shall be brought up equally on both sides of the pipe ensuring that it is in contact with the underside of the pipe barrel and be carefully compacted in layers not exceeding 150 mm thickness ensuring full compaction next to the trench walls.

3.4 UTILITY TRENCH BACKFILL

- A. Do not begin backfilling until pipe systems are inspected, approved and comply with applicable codes.
- B. Material indicated for pipe bedding may be substituted for backfill material, from top of pipe bedding to 300 mm above top of pipe.
- C. Deposit backfill in maximum of 150 mm layers.
- D. Place utility warning and identification tape at a depth ~~between 200 mm to~~ of 300 mm below final grade ~~above the utility asset.~~
- ~~E. “Flowable Fill” trench backfill:~~

- ~~1. Use straps or anchors as required to maintain proper alignment at pipe. Care shall be taken to prevent movement or floatation of pipe during fill placement and intrusion of "Flowable Fill" into undesirable or unintended locations.~~
- ~~2. "Flowable Fill" shall not be placed within 450 mm of finished grade.~~
- ~~3. "Flowable Fill" shall not be subsequently covered until a minimum compressive strength of 205 kPa has been attained.~~

3.5 COMPACTION OF SOIL FILL

- A. Establish moisture-density relation of soils according to EN 1997. Moisten or aerate material to maintain its moisture content within -2 to +2 percent of optimum; do not puddle.
- B. Compact each layer of material to the following minimum percentages of maximum dry density when tested according to EN 1997-1:
 - 1. Under Footing: 100 percent (EN 1997-1).
 - 2. Under Slab-On-Grade: 98 percent (EN 1997-1).
 - 3. Under Exterior Paved Areas: 95 percent for top 300 mm horizontally 600 mm beyond edges of paving. 95 percent (EN 1997-1) for fills below top 150 mm.
 - 4. Lawn and Planted Areas: 92 percent (EN 1997-1) for fills 300 mm and deeper. 92 percent for fills in top 300 mm of depth.
 - 5. Pipe Bedding (outside building areas and not below pavements): 92 percent (EN 1997-1).
- C. Perform compaction by rolling with approved sheeps-foot or pad-foot rollers, or other equipment suitable to particular soil being compacted and size of area of Work. Do not use rollers or other heavy equipment closer than 1200 mm above Project Site piping systems, unless specifically approved in writing.
- D. Operate heavy equipment for compacting backfill sufficiently far enough away from walls to prevent surcharge damage, but in no case closer than 1000 mm. Compact backfill in areas remaining by means of power driven hand tampers or other approved equipment.
- E. As compaction of fill in each work area is completed, leave area undisturbed to allow for inspection and testing. Do not place additional fill material until area has been approved.
- F. Compact porous fill under slabs-on-grade to at least 70 percent relative density or 95 percent of dry density (EN 1997-1).

3.6 DISPOSAL OF EXCESS MATERIALS

- A. Excess material shall be stockpiled on Project Site as directed by Construction Manager. Protect stockpiled material with impermeable tarps or planting with temporary grass cover or perimeter silt fence to minimize erosion.
- B. Any material to be disposed of off-site shall be removed by the Contractor, who shall have a permit/license under the National Waste Management Legislation. Waste shall be deposited in a licensed waste reception facility in compliance with the National Waste Management Legislation and the Local Authority Waste Management plan.
- C. Explosives shall not be used and blasting on site is not permitted without written consent from the owner and approval from any associated local / regional administrative body.

Table 312323/A: Acceptable Earthworks Materials: Classification and Compaction Requirements									
Class	General Material Description	Typical Use	Permitted Constituents	Material Properties Required for Acceptability			Compaction Requirements		
				Property	Test in Accordance with	Acceptable Limits			
						LOWER		UPPER	
General Granular Fill	1A	Well graded granular material	General Fill	Any natural material or combination of natural materials. Recycled Aggregate	(i) grading	DS EN ISO 17892	Table 312323/B	Table 312323/B	Table 312323/D Method 2
					(ii) uniformity coefficient	See footnote 5	10	-	
					(iii) moisture content	DS EN ISO 17892	OMC – 3% (BS 2.5kg rammer)	OMC + 1% (BS 2.5kg rammer)	
					(iv) MCV	Note 5 below	8	15	
	1B	Uniformly graded granular material	General Fill	Any natural material or combination of natural materials Recycled Aggregate	(i) grading	DS EN ISO 17892	Table 312323/B	Table 312323/B	Table 312323/D Method 3
					(ii) uniformity coefficient	See footnote 5	-	10	
					(iii) moisture content	DS EN ISO 17892	OMC – 3% (BS 2.5kg rammer)	OMC + 1% (BS 2.5kg rammer)	
					(iv) MCV	Note 5 below	8	15	
	1C	Coarse granular material	General Fill	Any natural material or combination of natural materials Recycled Aggregate	(i) grading	DS EN ISO 17892	Table 312323/B	Table 312323/B	Table 312323/D Method 5
					(ii) uniformity coefficient	See footnote 11		-	
(iii) MCV					Note 5 below	8	15		
(iv) Los Angeles Coefficient					Note 5 below	-	50		

Table 312323/A: Acceptable Earthworks Materials: Classification and Compaction Requirements									
Class	General Material Description	Typical Use	Permitted Constituents	Material Properties Required for Acceptability				Compaction Requirements	
				Property	Test in Accordance with	Acceptable Limits			
General Cohesive Fill	2A Wet Cohesive Material	General Fill	Any natural material or combination of natural materials	(i) grading	DS EN ISO 17892	Table 312323/B	Table 312323/B	Table 312323/D Method 1 except for materials with a liquid limit greater than 50 as determined by DS EN ISO 17892, only tamping or grid rollers shall be used.	
				(ii) Plastic Limit (PL)	DS EN ISO 17892	-	-		
				(iii) moisture content	DS EN ISO 17892	PL – 4%	NR		
				(iv) MCV	Note 5 below	8	15		
				(i) grading	DS EN ISO 17892	Table 312323/B	Table 312323/B	Table 312323/D Method 2	
				(ii) Plastic Limit (PL)	DS EN ISO 17892	-	-		
				(iii) moisture content	DS EN ISO 17892	NR	PL – 4%		
2B Dry cohesive material	General Fill	Any natural material or combination of natural materials	(iv) MCV	Note 5 below	8	15	Table 312323/D Method 2		
			(i) grading	DS EN ISO 17892	Table 312323/B	Table 312323/B			
			(ii) moisture content	DS EN ISO 17892	NR	NR			
			(iii) MCV	Note 5 below	8	15			
2C Stoney cohesive material	General Fill	Any natural material or combination of natural materials	(i) grading	DS EN ISO 17892	Table 312323/B	Table 312323/B	Table 312323/D Method 2		
			(ii) moisture content	DS EN ISO 17892	NR	NR			
			(iii) MCV	Note 5 below	8	15			
			(iv) Undrained shear strength (remoulded)	DS EN ISO 17892	-	-			
2D Silty cohesive material	General Fill	Any natural material or combination of materials	(i) grading	DS EN ISO 17892	Table 312323/B	Table 312323/B	Table 312323/D Method 3		
			(ii) moisture content	DS EN ISO 17892	NR	NR			
			(iii) MCV	Note 5 below	8	15			
			(iv) Undrained shear strength (remoulded)	DS EN ISO 17892	-	-			
Landscape Fill	4 Various	Fill to Landscape Areas	Any natural material subject to specification approvals	(i) grading	DS EN ISO 17892	To be decided by PM on site	To be decided by PM on site		
				(ii) moisture content	DS EN ISO 17892	NR	NR		
				(iii) MCV	Note 5 below	6	18		

Table 312323/A: Acceptable Earthworks Materials: Classification and Compaction Requirements

Class	General Material Description	Typical Use	Permitted Constituents (Subject to Requirements of Clause 601 and Appendix 6(1))	Material Properties Required for Acceptability (In Addition to Requirements on Use of Fill Materials in Clause 601 and Testing in Clause 631)					Compaction Requirements
				Property	Test in Accordance with	Acceptable Limits			
Topsoil	5A	Topsoil, or turf existing on site	Topsoiling	Topsoil or turf designated as Class 5A in the contract	(i) grading	DS EN ISO 17892	-	DS EN ISO 17892	-
	5B	Imported Topsoil	Topsoiling	Natural Material complying with the specification	-	-	-	-	-
	5C	Imported Topsoil	Turfing	Natural Material complying with the specification	-	-	-	-	-
	Selected granular fill	6A	Select well graded granular material	Below water	Natural gravel, crushed gravel, crushed rock, recycled aggregate other than argillaceous rock or gravel, crushed concrete, or any combination thereof	(i) grading (ii) uniformity coefficient (iii) plasticity index	DS EN ISO 17892 See footnote 11 DS EN ISO 17892	Table 312323/B 10 Non-plastic	Table 312323/B - Non-plastic
6B		Select coarse granular material	Starter layer	Natural gravel, crushed rock, crushed concrete, recycled aggregate or any combination thereof	(i) grading (ii) plasticity index (iii) Los Angeles Coefficient	DS EN ISO 17892 See footnote 11 DS EN ISO 17892 Note 6 below	Table 312323/B Non-plastic -	Table 312323/B Non-plastic 50	Table 312323/D Method 5
6C		Select well graded granular material	Starter layer	Natural gravel, crushed gravel, crushed rock, recycled aggregate other than argillaceous rock or gravel, crushed concrete, or any combination thereof.	(i) grading (ii) uniformity coefficient (iii) plasticity index (iv) Los Angeles Coefficient (v) moisture content (vi) OMC (Vibrating Hammer)	DS EN ISO 17892 See footnote 5 DS EN ISO 17892 Note 6 below DS EN ISO 17892	Table 312323/B Non-plastic -	Table 312323/B Non-plastic 50 OMC - 2% (vibrating hammer)	Table 312323/D Method 3

Table 312323/A: Acceptable Earthworks Materials: Classification and Compaction Requirements									
Class	General Material Description	Typical Use	Permitted Constituents	Material Properties Required for Acceptability (In Addition to Requirements on Use of Fill Materials in Clause 601 and Testing in Clause 631)					Compaction Requirements
				Property	Test in Accordance with		Acceptable Limits		
							LOWER	UPPER	
Selected Granular fill continued	6E Selected granular material (Class 9A)	For stabilisation with cement to form capping	Any material, or combination of materials, other than unburnt colliery spoil and argillaceous rock. Properties (i), (ii) and (iii) in next column shall not apply to chalk.) Recycled aggregate	(i) Grading	DS EN ISO 17892 (on site)	Table 312323/B	Table 312323/B	Not applicable	
					DS EN 933-2 (off site)	Table 312323/B	Table 312323/B		
				(ii) mc	DS EN ISO 17892	-	To be decided by PM on site		
				(iii) liquid limit	DS EN ISO 17892	-	45		
				(iv) plasticity index	DS EN ISO 17892	-	20		
				(v) organic matter	DS EN ISO 17892	-	To be decided by PM on site		
				(vi) water soluble (WS) sulfate content	BS EN 1744-1 clause 10	-	3000 mg/l as so ₄		
				(vii) oxidisable sulphides (OS) content	BS EN 1744-1 clause 13		0.6% as so ₄		
				(viii) SMC of chalk	DS EN ISO 17892		20%		
				6F1	Selected granular material (fine grading)	Capping	Natural gravel, crushed gravel, recycled aggregate or combination of both, other than unburnt colliery spoil, argillaceous rock and chalk		(i) grading
(ii) optimum moisture content	DS EN ISO 17892	-	-					Method 6	
(iii) moisture content	DS EN ISO 17892	OMC-2% (vibrating hammer)	OMC (vibrating hammer)						
(iv) Los Angeles Coefficient	Note 6 below	-	60						
6F2	Selected granular material (coarse grading)	Capping	Crushed rock, recycled aggregate other than unburnt colliery spoil and argillaceous rock.						(i) grading
				(ii) optimum moisture content	DS EN ISO 17892	-	-	Method 6	
				(iii) moisture content	DS EN ISO 17892	OMC-2% (vibrating hammer)	OMC (vibrating hammer)		
				(iv) Los Angeles Coefficient	Note 6 below	-	50		

Table 312323/A: Acceptable Earthworks Materials: Classification and Compaction Requirements								
Class	General Material Description	Typical Use	Permitted Constituents	Material Properties Required for Acceptability			Compaction Requirements	
				Property	Test in Accordance with	Acceptable Limits		
						LOWER		UPPER
6N	Selected granular material	Fill to structures or upfill below structures	Crushed rock, other than argillaceous rock and crushed concrete	(i) grading	DS EN ISO 17892	Table 312323/B	Table 312323/B	End product: 95% of maximum dry density of DS EN ISO 17892 (vibrating hammer method)
				(ii) uniformity co-efficient	See Note 5	10	-	
				(iii) Los Angeles Coefficient	Note 6 below	-	40	
				(iv) undrained shear parameters (c & φ)	DS EN ISO 17892	NR	NR	
				(v) effective angle of internal friction (φ) & effective cohesion (c')	DS EN ISO 17892	φ = 35° c' = 0 kPa	NR	
				(vi) permeability	DS EN ISO 17892	5 x 10-5 m/s	-	
				(vii) moisture content	DS EN ISO 17892	OMC -2% (vibrating hammer)	OMC +2% (vibrating hammer)	
				(viii) MCV	Note 5 below	NR	NR	
				(ix) Slope stability test	DS EN ISO 17892			

Table 312323A: Acceptable Earthworks Materials: Classification and Compaction Requirements									
Class	General Material Description	Typical Use	Permitted Constituents	Material Properties Required for Acceptability				Compaction Requirements	
				Property	Test in Accordance with	Acceptable Limits			
Selected Granular fill	gp Selected granular material	Fill to structures or upfill below structures	Natural gravel, crushed gravel or combination of both excluding argillaceous gravel	(i) grading	DS EN ISO 17892	Table 312323B	Table 312323B	End product 95% of maximum dry density of DS EN ISO 17892 (vibrating hammer method)	
				(ii) uniformly co-efficient	See Note 5	5	-		
				(iii) Los Angeles Coefficient	Note 6 below	-	60		
				(iv) undrained shear parameters (c & ϕ)	DS EN ISO 17892	NR	NR		
				(v) effective angle of internal friction (ϕ) & effective cohesion (c')	DS EN ISO 17892	$\phi = 35^\circ$ c' = 0 kPa	NR		
				(vi) permeability	DS EN ISO 17892	5 x 10-5 m/s	-		
				(vii) moisture content	DS EN ISO 17892	OMC -2% (vibrating hammer)	OMC +2% (vibrating hammer)		
				(viii) MCV	Note 5 below	NR	NR		
				(ix) Slope stability test (where required)	DS EN ISO 17892				
				(i) Grading	DS EN ISO 17892 (on site) BS EN 933-2 (off site)	Table 312323B Table 312323B	Table 312323B Table 312323B	Not applicable	
6R Selected granular material	For stabilisation with lime and cement to form capping rock. Properties (i), (ii) and (iii) in next column shall not apply to chalk. Recycled aggregate			(ii) mc	DS EN ISO 17892	To be decided by PM on site	-		
				(iii) liquid limit	DS EN ISO 17892	-	45		
				(iv) plasticity index	DS EN ISO 17892	-	20		
				(v) organic matter	DS EN ISO 17892	-	To be decided by PM on site		

Table 312323/A: Acceptable Earthworks Materials: Classification and Compaction Requirements								
Class	General Material Description	Typical Use	Permitted Constituents	Material Properties Required for Acceptability			Compaction Requirements	
				Property	Test in Accordance with	Acceptable Limits LOWERUPPER		
Selected Cohesive Fill	Selected cohesive material	For stabilisation with lime to form capping (Class 9D)	Any material, or combination of materials, other than unburnt colliery spoil	(i) grading	DS EN ISO 17892	Table 312323/B	Table 312323/B	Not applicable
				(ii) mc	DS EN ISO 17892	-	NR	
				(iii) MCV	Note 5 below	NR	-	
				(iv) plasticity index	DS EN ISO 17892	10	-	
				(v) organic matter	DS EN ISO 17892	-	To be decided by PM on site	
				(vi) water soluble (WS) sulfate content	DS EN 1744-1 clause 10	-	To be decided by PM on site	
				(vii) oxidisable sulfides (OS) content	DS EN 1744-1 clause 13	-	To be decided by PM on site	
				(viii) total potential sulfate (TPS) contents	DS EN 1744-1 clause 11	-	To be decided by PM on site	

Table 312323/A: Acceptable Earthworks Materials: Classification and Compaction Requirements								
Class	General Material Description	Typical Use	Permitted Constituents	Material Properties Required for Acceptability			Compaction Requirements	
				Property	Test in Accordance with	Acceptable Limits		
Selected Cohesive Fill	Selected silty cohesive material	For stabilisation with cement to form capping (Class 9B)	Any material, ore combination of materials, other than chalk, unburnt colliery spoil and argillaceous rock	(i) grading	DS EN ISO 17892	Table 312323/B	Table 312323/B	Not applicable
				(ii) uniformity coefficient	See Note 5	5	-	
				(iii) mnc	DS EN ISO 17892	-	NR	
				(iv) MCV	Note 5 below	NR	-	
				(v) liquid limit	DS EN ISO 17892	-	45	
				(vi) plasticity index	DS EN ISO 17892	-	20	
				(vii) organic matter	DS EN ISO 17892	-	To be decided by PM on site	
				(viii) water soluble (WS) sulfate content	DS EN 1744-1 clause 10	-	To be decided by PM on site	
				(ix) oxidisable sulfides (OS) content	DS EN 1744-1 clause 13	-	To be decided by PM on site	
				(x) total potential sulfate (TPS) contents	DS EN 1744-1 clause 11	-	To be decided by PM on site	
				(i) grading	DS EN ISO 17892	Table 312323/B	Table 312323/B	
				(ii) mc	DS EN ISO 17892	-	NR	
				(iii) MCV	Note 5 below	NR	-	
				(iv) plasticity index	DS EN ISO 17892	10	-	
71	Selected cohesive material	For stabilisation with lime and cement to form capping (Class 9E)	Any material, or combination of materials, other than unburnt colliery spoil	(v) organic matter	DS EN ISO 17892	-	To be decided by PM on site	
				(vi) water soluble (WS) sulfate content	DS EN 1744-1 clause 10	-	To be decided by PM on site	
				(vii) oxidisable sulfides (OS) content	DS EN 1744-1 clause 13	-	To be decided by PM on site	
				(viii) total potential sulfate (TPS) contents	DS EN 1744-1 clause 11	-	To be decided by PM on site	

Table 312323/A: Acceptable Earthworks Materials: Classification and Compaction Requirements							
Class	General Material Description	Typical Use	Permitted Constituents	Material Properties Required for Acceptability			Compaction Requirements
				Property	Test in Accordance with	Acceptable Limits LOWER UPPER	
Stabilised materials (Refer Footnote 12)	Cement stabilised well graded granular material	Capping	Class 6E with addition of cement according to 03300	(i) pulverisation	DS EN 13286	60%	-
				(ii) bearing ratio	DS EN 13286	15	-
				(iii) mc	DS EN 13286	To enable compaction to this specification	
						density of DS EN 13286(2.5kg rammer method)	
9A	Cement stabilised silty cohesive material	Capping	Class 7F with addition of cement according to 03300	(i) pulverisation	DS EN 13286	60%	-
				(ii) MCV immediately before compaction	Note 5 below	8.5	12
				(iii) bearing ratio	DS EN 13286	15	-
				(iv) mc	DS EN 13286	To enable compaction to this specification	
9B	Cement stabilised silty cohesive material	Capping	Class 7F with addition of cement according to 03300	(i) pulverisation	DS EN 13286	60%	-
				(ii) bearing ratio	DS EN 13286	15	-
				(iii) mc	DS EN 13286	To enable compaction to this specification	
						density of DS EN 13286(2.5kg rammer method)	
9C	Cement stabilised conditioned pulverised fuel ash cohesive material	Capping	Class 7G with addition of cement according to 03300	(i) pulverisation	DS EN 13286	60%	-
				(ii) bearing ratio	DS EN 13286	15	-
				(iii) mc	DS EN 13286	To enable compaction to this specification	
						density of DS EN 13286(2.5kg rammer method)	

Footnotes to Table 312323/A

1. App: Appendix
2. Tab: Table
3. NR: Not required
4. Uniformity coefficient is defined as the ratio of the particle diameters
5. D60 to D10 on the particle size distribution curve where:
 - a. D60: Particle diameter at which 60 percent of the soil by weight is finer
 - b. D10: Particle diameter at which 10 percent of the soil by weight is finer
6. Where the Moisture Condition Value (MCV) is to be determined, the determination shall be carried out in accordance with EN ISO 17892.
7. The determination of the MCV/moisture content relation in accordance with EN ISO 17892 shall be carried out when required.
8. Los Angeles and Other Tests for Particle Soundness
9. Resistance to Fragmentation – Los Angeles Coefficient (LA)
10. The value of Los Angeles coefficient shall be determined in accordance with EN 1097-2.
11. Other Tests for Particle Soundness
12. Where magnesium sulfate soundness tests are required to be carried out, they shall be carried out in accordance with EN 1367-2. Where slake durability tests are required, point load or other tests for soundness to be carried out.

Table 312323/B- Grading Requirements for Acceptable Earthworks Materials																					
Percentage by Mass Passing the Size Shown																					
Class	Size (mm)		Size (mm) DS EN 933										Size (microns)					Size (microns)	Class		
	500	300	125	90	75	37.5	28	20	14	10	6.3	5	3.35	2	1.18	600	300			150	63
	500	300	125	90	75	37.5	28	20	14	10	6.3	5	3.35	2	1.18	600	300	150	63	2	
1A		100	95-100																<15		1A
1B			100																<15		1B
1C	100		10-95													0-25			15		1C
2A & 2B			100											80-100					15-100		2A & 2B
2C			100											15-80					15-80		2C
2D			100																80-100	0-20	2D
6A	100									0-100		0-85				0-45			0-5		6A
6B	100		0-10																		6B
6C			100				0-100				0-100		0-35	0-10		0-2					6C
6D										100		89-100		60-100	30-100	15-80	5-48	0-15 except 0-20 for crushed rock			6D
6E & 6R			100	85-100						25-100						10-100				<15	6E & 6R

Table 312323/B- Grading Requirements for Acceptable Earthworks Materials																					
Percentage by Mass Passing the Size Shown																					
Class	Size (mm)		Size (mm) DS EN 933												Size (microns)				Size (microns)	Class	
	500	300	125	90	75	37.5	28	20	14	10	6.3	5	3.35	2	1.18	600	300	150			63
6F1					100	75-100				40-95		30-85					10-50			<15	6F1
6F2			100	80-100	65-100	45-100				15-60		10-45				0-25			0-12	6F2	
6F3			100	80-100	65-100	45-100				15-60		10-45				0-25			0-12	6F3	
6H								100				60-100			15-45	0-25		0-5		6H	
6I & 6J			100		85-100				25-100					15-100		9-100			<15	6I & 6J	
6K								100											0-10	6K	
6L										100		89-100		60-100	30-100	15-100	5-70	0-15 except 0-20 for crushed rock		6L	

Table 312323/C Limits of material properties of fill for use with metal components in reinforced soul and anchored earth structures for Class 6H, 6I, 6J, 7C and 7D materials

Reinforcing Element Material	Properties of Fill								
	pH Value		Max Chloride Ion Content %	Max Organic Content %	Max Water Soluble (WS) Sulfate Content mg/l as SO ₄	Maximum Oxidisable Sulfides (OS) Content % as SO ₄	Minimum Restivity Ohm.cm	Minimum Redox Potential volts	Microbial Activity Index
	Min	Max							
Galvanised Steel	5	10	0.02	0.2	300	0.06	5000	0.40	)
Stainless Steel	5	10	0.025	0.2	600	0.12	3000	0.35	) Less than 5
									)

NOTES:

- A. A method of calculating the Microbial Activity Index may be obtained by reference to TRRL Contractor Report 54 "Soil Corrosivity Assessment."
- B. The corrosion potential of frictional fill shall be assessed from resistivity, pH, chloride, water soluble sulfate and oxidisable sulfides tests. For cohesive soil it will be necessary to test additionally for organic content. Should either organic content or sulfate be in excess of the specified levels, then tests shall also be included for Redox Potential and Microbial Activity Index.
- C. The water soluble sulfate content and oxidisable sulfides content shall be determined in accordance with the methods described in EN 1744-1 clause 10, 11 and 13.
- D. Methods of test (except for Microbial Activity Index, water soluble sulfate content and oxidisable sulfides content) are given in EN ISO 17892.

Table 312323/D Method Compaction for earthworks Materials: Plant and Methods (Method 1 to Method 6) Compaction shall be either method or end-product as required for the Class of fill in Table 6/1, using plant appropriate to the Class of fill and the site conditions

Type of Compaction Plant	Ref No.	Category	Method 1		Method 2		Method 3		Method 4		Method 5		Method 6		
			D	N#	D	N#	D	N#	D	N	D	N	N for D = 110 mm	N for D = 150 mm	N for D = 250 mm
Smoothed wheeled roller (or vibratory roller operating without vibration)	1	Mass per metre width of roll: over 2100 kg up to 2700 kg	125	8	125	10	125	10*	175	4	unsuitable	unsuitable	unsuitable	unsuitable	unsuitable
	2	over 2700 kg up to 5400 kg	125	6	125	8	125	8*	200	4	unsuitable	unsuitable	16	unsuitable	unsuitable
	3	over 5400 kg	150	4	150	8	unsuitable	unsuitable	300	4	unsuitable	unsuitable	8	16	unsuitable
Grid roller	1	Mass per metre width of roll: over 2700 kg up to 5400 kg	150	10	unsuitable	unsuitable	150	10	250	4	unsuitable	unsuitable	unsuitable	unsuitable	unsuitable
	2	over 5400 kg up to 8000 kg	150	8	125	12	unsuitable	unsuitable	325	4	unsuitable	unsuitable	20	unsuitable	unsuitable
	3	over 8000 kg	150	4	150	12	unsuitable	unsuitable	400	4	unsuitable	unsuitable	12	20	unsuitable
Deadweight tamping roller	1	Mass per metre width of roll: over 4000 kg up to 6000 kg	225	4	150	12	250	4	350	4	unsuitable	unsuitable	12	20	unsuitable
	2	over 6000 kg	300	5	200	12	300	3	400	4	unsuitable	unsuitable	8	12	20
Pneumatic-tyred roller	1	Mass per wheel: over 1000 kg up to 1500 kg	125	6	unsuitable	unsuitable	150	10*	240	4	unsuitable	unsuitable	unsuitable	unsuitable	unsuitable
	2	over 1500 kg up to 2000 kg	150	5	unsuitable	unsuitable	unsuitable	unsuitable	300	4	unsuitable	unsuitable	unsuitable	unsuitable	unsuitable
	3	over 2000 kg up to 2500 kg	175	4	125	12	unsuitable	unsuitable	350	4	unsuitable	unsuitable	unsuitable	unsuitable	unsuitable
	4	over 2500 kg up to 4000 kg	225	4	125	10	unsuitable	unsuitable	400	4	unsuitable	unsuitable	unsuitable	unsuitable	unsuitable
	5	over 4000 kg up to 6000 kg	300	4	125	10	unsuitable	unsuitable	unsuitable	unsuitable	unsuitable	unsuitable	12	unsuitable	unsuitable
	6	over 6000 kg up to 8000 kg	350	4	150	8	unsuitable	unsuitable	unsuitable	unsuitable	unsuitable	unsuitable	12	unsuitable	unsuitable
	7	over 8000 kg up to 12000 kg	400	4	150	8	unsuitable	unsuitable	unsuitable	unsuitable	unsuitable	unsuitable	10	16	unsuitable
	8	over 12000 kg	450	4	175	6	unsuitable	unsuitable	unsuitable	unsuitable	unsuitable	unsuitable	8	12	unsuitable
Vibratory tamping roller	1	Mass per metre width of a vibrating roll: over 700 kg up to 1300 kg	100	12	100	12	150	12	100	10	unsuitable	unsuitable	unsuitable	unsuitable	unsuitable
	2	over 1300 kg up to 1800 kg	125	12	125	12	175	12*	175	8	unsuitable	unsuitable	12	unsuitable	unsuitable
	3	over 1800 kg up to 2300 kg	150	12	150	12	200	12*	unsuitable	unsuitable	unsuitable	unsuitable	8	12	unsuitable
	4	over 2300 kg up to 2900 kg	150	9	150	9	250	12*	unsuitable	unsuitable	400	5	6	10	unsuitable
	5	over 2900 kg up to 3600 kg	200	9	200	9	275	12*	unsuitable	unsuitable	500	6	6	10	unsuitable
	6	over 3600 kg up to 4300 kg	225	9	225	9	300	12*	unsuitable	unsuitable	600	6	4	8	unsuitable
	7	over 4300 kg up to 5000 kg	250	9	250	9	300	9*	unsuitable	unsuitable	700	6	3	7	12
	8	over 5000 kg	275	9	275	9	300	7*	unsuitable	unsuitable	800	6	3	6	10

Type of Compaction Plant	Ref No.	Category	Method 1			Method 2			Method 3			Method 4			Method 5			Method 6		
			D	N#		D	N#		D	N#		Δ	N		D	N		N for D = 110 mm	N for D = 150 mm	N for D = 250 mm
Vibratory roller		Mass per metre width of vibratory roll:																		
	1	over 270 kg up to 450 kg	unsuitable			75	16	150	16		unsuitable				unsuitable			unsuitable		unsuitable
	2	over 450 kg up to 700 kg	unsuitable			75	12	150	12		unsuitable				unsuitable			unsuitable		unsuitable
	3	over 700 kg up to 1300 kg	100	12		125	10	150	6		125	10			unsuitable			16	unsuitable	unsuitable
	4	over 1300 kg up to 1800 kg	125	8		150	8	200	10*		175	4			unsuitable			6	16	unsuitable
	5	over 1800 kg up to 2300 kg	150	4		150	4	225	12*		unsuitable				unsuitable			4	6	12
	6	over 2300 kg up to 2900 kg	175	4		175	4	250	10*		unsuitable				400	5		3	5	11
	7	over 2900 kg up to 3600 kg	200	4		200	4	275	8*		unsuitable				500	5		3	5	10
	8	over 3600 kg up to 4300 kg	225	4		225	4	300	8*		unsuitable				600	5		2	4	8
	9	over 4300 kg up to 5000 kg	250	4		250	4	300	6*		unsuitable				700	5		2	4	7
	10	over 5000 kg	275	4		275	4	300	4*		unsuitable				800	5		2	3	6
Vibrating plate compactor		Mass per m ² of base plate:																		
	1	over 880 kg up to 1100 kg	unsuitable			unsuitable		75	6		unsuitable				unsuitable			unsuitable		unsuitable
	2	over 1100 kg up to 1200 kg	unsuitable			75	10	100	6		75	10			unsuitable			unsuitable		unsuitable
	3	over 1200 kg up to 1400 kg	unsuitable			75	6	150	6		150	8			unsuitable			unsuitable		unsuitable
	4	over 1400 kg up to 1800 kg	100	6		125	6	150	4		unsuitable				unsuitable			8	unsuitable	unsuitable
	5	over 1800 kg up to 2100 kg	150	6		150	5	200	4		unsuitable				unsuitable			5	8	unsuitable
	6	over 2100 kg	200	6		200	5	250	4		unsuitable				unsuitable			3	6	12
Vibro-tamper		Mass:																		
	1	over 50 kg up to 65 kg	100	3		100	3	150	3		125	3			unsuitable			4	8	unsuitable
	2	over 65 kg up to 75 kg	125	3		125	3	200	3		150	3			unsuitable			3	6	12
	3	over 75 kg up to 100 kg	150	3		150	3	225	3		175	3			unsuitable			2	4	10
	4	over 100 kg	225	3		200	3	225	3		250	3			unsuitable			2	4	10
Power rammer		Mass:																		
	1	100 kg up to 500 kg	150	4		150	6	unsuitable			200	4			unsuitable			5	8	unsuitable
	2	over 500 kg	275	8		275	12	unsuitable			400	4			unsuitable			5	8	14
Dropping-weight compactor		Mass of rammer over 500 kg																		
	1	weight drop: over 1 m up to 2 m	600	4		600	8	450	8		unsuitable				unsuitable			unsuitable		unsuitable
	2	over 2 m	600	2		600	8	unsuitable			unsuitable				unsuitable			unsuitable		unsuitable

Table 312323/D

Type of Compaction Plant	Ref~ No.	Category	Method 7	
			N for D = 150 mm	N for D = 250mm
Smooth wheeled roller (or vibratory roller operating without vibration)	1 2 3	Mass per metre width of roll: over 2100 kg up to 2700 kg over 2700 kg up to 5400 kg over 5400 kg	unsuitable unsuitable 12	unsuitable unsuitable unsuitable
Grid roller	1 2 3	Mass per metre width of roll: over 2700 kg up to 5400 kg over 5400 kg up to 8000 kg over 8000 kg	unsuitable 16 8	unsuitable unsuitable unsuitable
Deadweight tamping roller	1 2	Mass per metre width of roll: over 4000 kg up to 6000 kg over 6000 kg	4 3	8 6
Pneumatic-tyred roller	1 2 3 4 5 6 7 8	Mass per wheel: over 1000 kg up to 1500 kg over 1500 kg up to 2000 kg over 2000 kg up to 2500 kg over 2500 kg up to 4000 kg over 4000 kg up to 6000 kg over 6000 kg up to 8000 kg over 8000 kg up to 12000 kg over 12000 kg	unsuitable 12 6 5 4 unsuitable unsuitable unsuitable	unsuitable unsuitable unsuitable unsuitable 16 8 4 4
Vibratory tamping roller	1 2 3 4 5 6 7 8	Mass per metre width of vibrating roll: over 700 kg up to 1300 kg over 1300 kg up to 1800 kg over 1800 kg up to 2300 kg over 2300 kg up to 2900 kg over 2900 kg up to 3600 kg over 3600 kg up to 4300 kg over 4300 kg up to 5000 kg over 5000 kg	unsuitable unsuitable 16 12 10 8 7 6	unsuitable unsuitable unsuitable unsuitable unsuitable 16 14 12
Vibratory roller	1 2 3 4 5 6 7 8 9 10	Mass per metre width of vibrating roll: over 270 kg up to 450 kg over 450 kg up to 700 kg over 700 kg up to 1300 kg over 1300 kg up to 1800 kg over 1800 kg up to 2300 kg over 2300 kg up to 2900 kg over 2900 kg up to 3600 kg over 3600 kg up to 4300 kg over 4300 kg up to 5000 kg over 5000 kg	unsuitable unsuitable unsuitable unsuitable 12 10 10 8 8 6	unsuitable unsuitable unsuitable unsuitable unsuitable unsuitable unsuitable unsuitable unsuitable 12
Vibratory plate compactor	1 2 3 4 5 6	Mass per m ² of base plate: over 880 kg up to 1100 kg over 1100 kg up to 1200 kg over 1200 kg up to 1400 kg over 1400 kg up to 1800 kg over 1800 kg up to 2100 kg over 2100 kg	unsuitable unsuitable unsuitable 10 8 6	unsuitable unsuitable unsuitable unsuitable unsuitable unsuitable
Vibro-tamper	1 2 3 4	Mass: over 50 kg up to 65 kg over 65 kg up to 75 kg over 75 kg up to 100 kg over 100 kg	unsuitable unsuitable unsuitable 8	unsuitable unsuitable unsuitable unsuitable
Power rammer	1 2	Mass: 100 kg up to 500 kg over 500 kg	8 6	unsuitable 10
Dropping weight compactor	1 2	Mass of rammer over 500 kg height drop: over 1 m up to 2 m over 2 m	unsuitable unsuitable	unsuitable unsuitable

Material Class	Requirement	Suggested Frequency Per Source
1 General Granular Fill	Grading/Uniformity Coeff	Twice a week
	mc/MCV	2 tests per 1000m ³ of material up to a max 5 per day
	SMC of Chalk	Twice a week
2 General Cohesive Fill	Grading	Twice a week
	mc/MCV/PL/Shear strength	2 tests per 1000m ³ of material up to max of 5 per day
	SMC	Twice a week
	Bulk Density (PFA)	2 tests per 1000m ³ of material up to max of 5 per day.
3 General Chalk Fill	mc	2 tests per 1000m ³ of material up to max 5 per day Fill of material
	SMC	Daily
4 Landscape Fill	Grading/mc/MCV	Daily
5 Topsoil	Grading	Daily
6 Selected Granular Fill	Grading/Uniformity Coeff	1 test per 400 tonnes of material
	PI/LL	Daily
	10% Fines Value/SMC	Weekly
	omc/mc/MCV	1 test per 400 tonnes of material
	Organic Matter/Total Sulphate Content	As required or weekly
	pH/Chloride Ion Content	As required or weekly
	Resistivity	As required
	Undrained Shear Parameters	As required
7	Grading/mc/MCV	1 test per 400 tonnes of material

Selected Cohesive Fill	SMC	Twice a week
	PI/LL	As required or daily
	Organic Matter/Total Sulphate Content	As required or weekly
	Resistivity	As required
	Undrained Shear Parameters	As required
	Permeability	As required
	Coeff of Friction/Adhesion	As required
8 Miscellaneous Fill	mc/MCV	Daily
9 Stabilised General Fill Materials	Pulverisation	1 test per lane width per 100m length
	mc/MCV	1 test per lane width per 100m length
	Bearing Ratio	1 test per lane width per 100m length

Compliance Testing of Placed Materials and Frequency of Tests

END OF SECTION

REVISION HISTORY

Version	Date	Description and location of change
0.1	August 2024	Initial version – BOD Draft submission
0.2	September 2024	BOD Final Submission
0.3	February 2025	Draft IFC Updates
1.0	12 th of March 2025	Final IFC Issue